Boolean algebras as lattices

We begin with the definition of a Boolean algebra viewed as an *ordered structure*. We shall see later that there is an alternate, equivalent definition of a Boolean algebra, viewed as *rings*.

Recall that a *partial order* $\leq$ on a set B is a binary relation that is *reflexive* $(x \leq x,\ x \in B)$, *transitive* (if $x \leq y$ and $y \leq z$ then $x \leq z$, $x, y, z \in B$) and *antisymmetric* (if $x \leq y$ and $y \leq x$ then $x = y$, $x, y \in B$).

$(B, \leq)$ forms a *lattice*, provided

(a) $(B, \leq)$ is a partially ordered set,

(b) any two elements x and y of B have a least upper bound $x \vee y$ and a greatest lower bound $x \wedge y$.

We denote the lattice by the tuple $(B, \leq, \vee, \wedge)$.

If the operations $\wedge$ and $\vee$ in the lattice distribute over each other, the lattice is said to be *distributive*.

If there is a least element 0 and a greatest element 1 in a lattice, it is said to be *bounded*, and moreover,

if every element x of a bounded lattice has a unique complement x^c such that $x \vee x^c = 1$ and $x \wedge x^c = 0$, the lattice is called *complemented*.

Definition 0.1. (**Boolean algebra**) A Boolean algebra $(B, \leq, \vee, \wedge,^c, 0, 1)$ is a complemented distributive lattice.

1. A prototypical example of a Boolean algebra (as will be substantiated in the sequel), called the *power set Boolean algebra*, is provided by the structure
   $(\mathcal{P}(U), \cup, \cap,^c, \emptyset, U)$
   on the power set $\mathcal{P}(U)$ of any non-empty set U.
   $\cup$ denotes the operation of union, $\cap$ that of intersection and ^{c} that of complement in $\mathcal{P}(X)$.

2. Consider the set $\mathcal{F}/\equiv$ with operations $\vee, \wedge, \neg$, along with the distinguished elements 1 and 0.
   $(\mathcal{F}/\equiv, \vee, \wedge, \neg, 0, 1)$ is a Boolean algebra.

Equivalently, we have the Boolean algebra
$(\mathcal{F}/\sim, \leq, \vee, \wedge, \neg, 0, 1)$.
Observe that $\mathcal{F}/\sim$ forms a Boolean algebra with the operations $\wedge, \vee, \neg$ and with theorems and contradictions of PL constituting the classes 1 and 0 respectively; all of this can be derived *directly* – without any reference to logical equivalence, using the properties of $\vdash$.

This algebra is called the *Lindenbaum-Tarski algebra* for PL.

3. We also have the Boolean algebra on the set $\mathbf{2}{:=}\ \{0,1\}$, viz. the structure $(\{0,1\}, \leq, \vee, \wedge, \neg, 0, 1)$, or equivalently, $(\{F,T\}, \leq, \vee, \wedge, \neg, F, T)$, where F, T are the two truth values in PL and $\vee, \wedge, \neg$ are the operations defining the truth tables.

We shall see later that the Lindenbaum-Tarski algebra and the Boolean algebra on $\mathbf{2}$ play an important role in giving a close connection between PL and the class of all Boolean algebras.

Note. There is also the *degenerate* Boolean algebra over a singleton, but we shall not consider it in this discourse.

Exercise 0.1. Prove that in each of the above examples, the structure mentioned is indeed a Boolean algebra.

Some simple properties:

Exercise 0.2.
(i) The glb operation ($\wedge$) and lub operation ($\vee$) are associative and commutative.

(ii) $x \vee 0 = x,\ x \wedge 0 = 0,\ x \vee 1 = 1$, and $x \wedge 1 = x$.

(iii) Any non-empty finite subset $\{x_1, \ldots, x_k\}$ of B has a glb and a lub in B.

(iv) $x \leq y,\ u \leq v$ imply $x \vee u \leq y \vee v$ as well as $x \wedge u \leq y \wedge v$.

We now list some properties of the complement of any element in a Boolean algebra.

Proposition 0.1.
(i) De Morgan's laws: $(x \wedge y)^c = x^c \vee y^c$; $(x \vee y)^c = x^c \wedge y^c$.

(ii) $x \leq y$ if and only if $y^c \leq x^c$.

(iii) $x \leq y^c$ if and only if $x \wedge y = 0$.

Proof. Proof is left as an exercise. □